MIL-W-50275A (MU)
15 October 1972
Superseding
MIL-W-50275 (MU)
10 April 1969

MILITARY SPECIFICATION

WARHEAD, HE, M18A1
PARTS FOR

1. SCOPE

1.1 This specification contains special requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembling and packing of one type of high explosive, shaped charge warhead designated as Warhead, HE, M18A1 and is intended for use in the Rocket, HE, 66MM, Anti-Tank, M72A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-D-1000 - Drawings, Engineering and Associated Lists

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

FSC: 1340

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

PICATINNY ARSENAL

9220011 - Warhead Assembly Metal Parts.

9251978 - Cap, Base Connector

PUBLICATIONS

MIL-HDBK - 204 - Inspection Equipment Design

PICATINNY ARSENAL

ET-9220011 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. *REQUIREMENTS*

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. - All parts used in the fabrication of the warhead shall comply with all requirements of all of the applicable drawings and specifications.

3.3 Assembly. - The warhead assembly shall comply with all of the requirements specified in drawing (dwg.) 9220011, and with all of the requirements specified in applicable specifications.

3.4 Identification of molded parts. - The molds used in the forming of molded components shall provide a durable identifying mark on a surface of the molded component which will identify the mold (and mold cavity) that formed the component. This identifying mark shall be located where it will not become obliterated, removed in subsequent finishing operations, and if practicable, be visible after assembly.

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3.5 Dielectric strength of flexible printed circuit. - When a one (1) inch portion of the flexible printed circuit is compressed between two electrically conductive plates with a minimum load of 1.7 pounds, placed in an ambient temperature of 190 degrees Fahrenheit (°F) minimum for a minimum of 25 minutes (while under compression), the leakage current shall not exceed 20 microamperes while an alternating current of 1,500 volts (root mean square) 60 cycles per second is applied at a rate of 500 volts per second minimum between both conductors and the compressing plates.

3.6 Lamination of flexible printed circuit. - There shall be no unbonded sections in the lamination of the flexible printed circuit when viewed under seven (7) power magnification.

3.7 Assembly force of base of the nose cap connector to the ogive. - The base (see dwg. 9278721) shall assemble to a fixture representing the nominal opening in the ogive and cross section form of the opening with a maximum force of 10 pounds. The base shall not break or crack during assembly and shall be capable of being removed without breaking or cracking and shall withstand a disassembly force of 2 pounds minimum.

3.8 Assembly of flexible cable assembly to base connector asembly. - The fuze end of the flexible cable assembly shall assemble to a base connector assembly 9251979 with a force of five (5) pounds minimum and twenty (20) pounds maximum. Both socket connectors shall assemble at the same time and shall seat fully to the fixture and shall not exhibit cracked solder joints, cracked or delaminated cable when viewed under seven (7) power magnification. The electrical resistance between each pin of the base connector assembly and the contact in the nose cap connector shall not exceed 0.5 ohms.

3.9 Leakage of explosive charge cavity. - The explosive charge cavity in the completed warhead parts assembly shall not contain any openings which, when considered cumulatively, shall not permit the leakage of air in excess of that of a 0.002 inch diameter hole in an otherwise sound explosive charge cavity.

3.10 Electrical continuity. - The electrical resistance indicative of acceptable electrical continuity shall not exceed 0.5 ohms measured between each contact (9278719) and the appropriate socket connector (9251970) in the complete warhead parts assembly.

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3.11 Dielectric strength of flexible cable assembly in complete warhead assembly. - The leakage current shall not exceed 20 microamperes while a minimum electrical test potential of 700 volts of direct current is applied between both conductors and between each conductor (or both conductors concurrently) and the ogive, body and liner.

3.12 Workmanship. - All soldered joints shall be free from scratches, roughness, sharp edges, dullness, looseness, blistering, foreign matter, and all other evidence of poor workmanship that will render the assemblies unsuitable for the purposes intended. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat, clean and sharply defined.

3.13 First article inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of:

125 contacts as per dwg. 9278719 from each supplier

125 socket, connectors as per dwg. 9251970 from each supplier

125 base, nose cap connector as per dwg. 9278721 from every mold cavity

125 flexible printed circuits as per dwg. 9251968 from each supplier

125 cover, nose cap connector assembly as per dwg. 9278722 from every mold cavity

125 retainer, nose cap connector assembly as per dwg. 9278720 from every mold cavity

125 cap, base connector as per dwg. 9251978 from every mold cavity

125 flexible cable assembly as per dwg. 9251980 from each supplier of flexible printed circuits

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10 liners as per dwg. 9220009

10 ogives as per dwg. 9220012

10 bodies as per dwg. 9220010

10 body assemblies as per dwg. 9242585

10 warhead assemblies metal parts as per dwg. 9220011

The product included in the first article sample shall be product that has been produced by the contractor or furnished by a supplier and has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials, (see 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Each component, sub-assembly and assembly in the first article sample shall be examined and inspected for every dimension and requirement of the applicable drawing, especially for the characteristics listed in 4.3.2 and then tested as specified in 4.3.3.

4.2.3 Rejection. - The first article sample shall be rejected if any component, sub-assembly or assembly fails to conform to the applicable requirements. The sample shall be rejected if any portion of the first article sample is classed defective when examined in accordance with 4.3.2 and 4.3.3. The Government reserves the right to terminate its inspection upon encountering any failure of the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample, or portion thereof. Until a first article sample is found acceptable, the contractor is in no way authorized by the Government to initiate regular production (including procurement of materials by the contractor) unless specifically directed by the Contracting Officer.

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4.3 Quality Conformance Inspection

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, General Inspection Level II, unless otherwise specified in the subparagraphs (adjacent to the AQL's). Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Cap, Base Connector (see dwg. 9251978).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 10.00 Percent-Inspection Level S-1		
101.	Overall dimensions oversize..	Comparator	01001
102.	Diameter of cavities, minimum	Gage	01002
103.	Location of cavities.....	Gage	01003
Minor:	AQL 1.00 Percent-Inspection Level II		
201.	Component incomplete or mold identification missing (See 3.4).....	Visual	01004
202.	Evidence of poor workmanship (see 3.12).....	Visual	01005

4.3.2.2 Cover, Nose Cap Connector Assembly (see dwg. 9278722 covering a detail of dwg. 9278723).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 10.00 Percent-Inspection Level S-1		
101.	Location of contact holes....	Gage	02001
102.	Diameter of contact holes, min.....	Gage	02002
Minor:	AQL 1.00 Percent-Inspection Level II		
201.	Component incomplete or mold identification missing (See 3.4).....	Visual	02003
202.	Evidence of poor workmanship (see 3.12).....	Visual	02004

4.3.2.3 Base, Nose Cap Connector Assembly (see dwg. 9278721 covering a detail of dwg. 9220011).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 10.00 Percent-Inspection Level S-1		
101.	Location of contact wells.....	Gage	03001
102.	Width of contact wells, minimum..	Gage	03002
103.	Diameter across retention legs, minimum.....	Gage	03003
Minor:	AQL 1.00 Percent-Inspection Level II		
201.	Component incomplete or mold identification missing (See 3.4).....	Visual	03004
202.	Evidence of poor workmanship (see 3.12).....	Visual	03005

4.3.2.4 Flexible Printed Circuit (see 3.6 and dwg. 9251968 covering a detail of dwg. 9251980).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Width, except at socket connector end, minimum.....	Gage	04001
102.	Thickness, minimum.....	Gage	04002
103.	Diameter of socket hole, maximum.	Gage	04003
104.	Delamination (see 3.6).....	7 Power Mag	04004
105.	Component incomplete.....	Visual	04005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	04006

4.3.2.5 Contact, Nose Cap Connector Assembly (see dwg. 9278719 covering a detail of dwg. 9278723).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Opening between contact prongs, maximum.....	Gage	05001
102.	Protective finish missing, damaged or incomplete exposing base metal.....	Visual	05002
103.	Component incomplete.....	Visual	05003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	05004

4.3.2.6 Socket Connector (see dwg. 9251970 covering a detail of dwg. 9251980).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Long inside diameter.....	Gage	06001
102.	Effective inside diameter between tabs (smallest inside diameter not including broached out portions.).....	Gage	06002
103.	Component incomplete.....	Visual	06003
104.	Protective finish missing, damaged or incomplete exposing base metal.....	Visual	06004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	06005

4.3.2.7 Assembly, Flexible Cable (see dwg. 9251980 covering a detail of dwg. 9242585).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Distance between socket connectors	Gage	07001
102.	Soldered connections defective (See 3.12).....	Visual	07002
103.	Assembly incomplete or damaged....	Visual	07003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.12).....	Visual	07004

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4.3.2.8 Liner (see dwg. 9220009 covering a detail of dwg. 9242585).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Wall variation exceeds drawing requirement between points A and B in any plane perpendicular to the longitudinal axis of the liner.	Gage	08001
102.	Length to large datum diameter.	Gage	08002
103.	Length to small datum diameter.	Gage	08003
104.	Diameter of body, head mating surface.....	Gage	08004
105.	Straightness between A and B...	Gage	08005
106.	Depth and width of slot.....	Gage	08006
107.	Wall thickness.....	Gage	08007
108.	Component split, scarred, pitted or dented.....	Visual	08008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	08009

4.3.2.9 Body (see dwg. 9220010 covering a detail of dwg. 9242585).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Wall thickness, mimimum.....	Gage	09001
102.	Outside diameter of small end..	Gage	09002
103.	Length of outside diameter of small end.....	Gage	09003
104.	Runout of small outside diameter to datum A.....	Gage	09004
105.	Straightness.....	Gage	09005
106.	Component cracked or damaged...	Visual	09006
Minor:	AQL 1.00 Percent		
201.	Protective finish missing, incompleted or damaged exposing base metal.....	Visual	09007
202.	Evidence of poor workmanship (see 3.12).....	Visual	09008

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4.3.2.10 Ogive (see dwg. 9220012 covering a detail of dwg. 9220011).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Wall thickness, minimum.....	Gage	10001
102.	Profile of small end including outside diameters of small end.	Gage	10002
103.	Inside diameter of small end, maximum.....	Gage	10003
Minor:	AQL 1.00 Percent		
201.	Protective finish missing, incomplete or damaged exposing base metal.....	Visual	10004
202.	Evidence of poor workmanship (see 3.12).....	Visual	10005

4.3.2.11 Assembly, Body (see dwg. 9242585 covering a detail of dwg. 9220011).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Concentricity of body with liner.....	Gage	11001
102.	Length from small end of body to first socket of flexible cable assembly.....	Gage	11002
103.	Excessive adhesive at liner and head juncture.....	Visual	11003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	11004

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4.3.2.12 Assembly, Warhead Metal Parts (see dwg. 9220011).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Ogive torque test failure.....	Test	12001
102.	Large outside diameter.....	Gage*	12002
103.	Runout of ogive to body.....	Gage	12003
104.	Flexible cable assembly improperly assembled or damaged.....	Visual	12004
105.	Paint missing, incomplete, or damaged (see 3.12).....	Visual	12005
106.	Marking missing, incomplete or illegible.....	Visual	12006
107.	Paint or other foreign matter in small outside body diameter	Visual	12007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	12008

* Outside diameter gage shall be at least as long as the length of the outside diameter of the assembly to assure a chamber gage effect.

4.3.3 Testing

4.3.3.1 First Article Sample.-The first article sample shall be examined 100 percent for all of the characteristics in 4.3.2, and then subjected to the test listed below, using test equipment in accordance with 4.3.4. Rejection of the first article sample shall be in accordance with 4.2.3.

4.3.3.1.1 Thickness of Protective Finish.-Contact, (See dwg. 9278719) - Code No. 05005; Socket Connector (See dwg. 9251970) Code No. 06006.-A sample of three (3) of each component shall be tested as specified in 4.4.1.

4.3.3.1.2 Dielectric Strength of Flexible Printed Circuit (See 3.5) - Code No. 04007.-The ten (10) flexible printed circuits shall be tested as specified in 4.4.2.

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4.3.3.1.3 Assembly Force of Base of the Nose Cap Connector to the Ogive (See 3.7) - Code No. 03006.-The ten (10) bases shall be tested as specified in 4.4.3.

4.3.3.1.4 Assembly Force of Flexible Cable Assembly to Case Connector Assembly (See 3.8) - Code No. 07005.-The ten (10) flexible cable assemblies shall be tested as specified in 4.4.4.

4.3.3.1.5 Leakage of Explosive Charge Cavity (See 3.9) - Code No. 12009.-The ten (10) complete warhead parts assemblies shall be tested as specified in 4.4.5.

4.3.3.1.6 Electrical Continuity; (See 3.10) - Code No. 12010.- The ten (10) complete warhead parts assemblies, after being tested as required in 4.3.3.1.5, shall be tested as specified in 4.4.6.

4.3.3.1.7 Dielectric Strength (See 3.11) - Code No. 12011.-The ten (10) complete warhead parts assemblies, after being tested as specified in 4.3.3.1.6, shall be tested as specified in 4.4.7.

4.3.3.2 *Production Testing*

4.3.3.2.1 Thickness of Protective Finish - Contact (See dwg. 9278719) - Major Defect - Code No. 05005; Socket Connector (See dwg. 9251970) - Major Defect - Code No. 06006.-A sample of three (3) components shall be selected from each plating batch and tested as specified in 4.4.1, using the test equipment specified in 4.3.4. The batch of components represented by the sample shall be rejected if one or more components are classed defective.

4.3.3.2.2 Dielectric Strength of Flexible Printed Circuit - (See 3.5) - Major Defect - Code No. 04007.-A sample of fifty (50) flexible printed circuits shall be selected from the lot and tested as specified in 4.4.2, using the test equipment specified in 4.3.4. The lot shall be rejected if two or more flexible printed circuits are classed defective.

4.3.3.2.3 Assembly Force of Base to Ogive (See 3.7) - Major Defect - Code No. 03006.-A sample of five (5) bases (See dwg. 9278721) shall be selected from each batch produced by each mold cavity and tested as specified in 4.4.3, using the test equipment specified in 4.3.4. The batch shall be rejected if one or more bases are classed defective.

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4.3.3.2.4 Assembly Force of Flexible Cable Assembly to Base Connector Assembly - (See 3.8) - Major Defect - Code No. 07005.- A sample of fifty (50) flexible cable assemblies shall be selected from the lot and tested as specified in 4.4.4, using the test equipment specified in 4.3.4. The lot shall be rejected if two (2) or more flexible cable assemblies are classed defective.

4.3.3.2.5 Leakage of Explosive Charge Cavity (See 3.9) - Major Defect - Code No. 12009.-This test shall be performed 100 percent. The lot shall be tested as specified in 4.4.5, using test equipment specified in 4.3.4. Defective assemblies shall be rejected and removed from the lot.

4.3.3.2.6 Electrical Continuity (See 3.10) - Major Defect - Code No. 12010.-This test shall be performed 100 percent. The lot of complete warhead metal parts assemblies shall be tested as specified in 4.4.6, using the test equipment specified in 4.3.4. Defective assemblies shall be rejected and removed from the lot.

4.3.3.2.7 Dielectric Strength (See 3.11) - Major Defect - Code No. 12011.-This test shall be performed 100 percent. The lot shall be tested as specified in 4.4.7, using the test equipment specified in 4.3.4. Defective assemblies shall be rejected and removed from the lot.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests required by this section is identified by Equipment Tabulation ET-9220011. (See 6.3).

4.4 Test Methods and Procedures

4.4.1 Thickness of Protective Finish.-The component shall be cross-sectioned through any convenient plane, and one cross-section polished and etched to enhance visible definition of the boundaries of the protective coating. The coating thickness shall be measured and the component classed defective if the thickness is less than the specified requirement.

4.4.2 Dielectric Strength of Flexible Printed Circuit.-The required portion of the flexible printed circuit shall be compressed by the specified load and placed in a chamber at the required temperature for the specified time. The test assembly shall be removed from the heat chamber and permitted to return to room temperature. The specified test voltage shall then be applied between the two conductors and the leakage current measured within 10 seconds after impressing the minimum required test voltage. The specified test voltage shall then be applied between each conductor (or both conductors concurrently), and both compressive plates and the leakage current measured within 10 seconds after impressing the minimum required test voltage. The test equipment shall provide a minimum of 100 microamperes of available current during the entire test. The flexible printed circuit shall be classed defective if any of the leakage current measurements exceed the specified requirement.

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4.4.3 Assembly Force of Base to Ogive.-The test equipment shall represent the small end of the ogive and the opening shall represent the mid-range of the dimensional extremes. The equipment shall use a dead load to assemble the base to the fixture, and a dead load to attempt removal from the fixture. Assembly of the base may be initiated by compressing the retentive legs and starting the legs into the simulated ogive opening. The dead load shall then be applied, and observation shall be made for complete assembling. The removal force or dead load to remove the base shall be applied and observation made of result. The base shall be examined for cracks or broken segments. The base shall be classed defective if it fails to fully assemble, or if it is withdrawn by the specified force, or if the base is cracked or broken.

4.4.4 Assembly Force of Flexible Cable Assembly to Base Connector Assembly.-The base connector assembly shall be assembled in the test fixture specified in 4.3.4. Place the fuze end of the cable assembly without the cap, base connector over the base connector assembly so that the two sockets of the cable assembly are placed over the two pins of the connector assembly. Using a regulated load and a fixture simulating the cap base connector, apply the load evenly and without shock over the two sockets of the cable assembly. Record maximum load required to assembly the two parts together. Measure resistance of each wire to the pin of the base connector. Test equipment, test and calibration procedures must be approved by Picatinny Arsenal prior to use.

4.4.5 Leakage of Explosive Cavity.-The warhead assembly shall be secured to the required test equipment, and a pressure differential introduced between the explosive cavity and all external surfaces. The test equipment shall be such that when a rejection test standard is tested by an established pressure level differential over a constant minimum time interval, both of which are maintained throughout testing, a consistent rejection signal indication results. Any warhead assembly which permits a leakage equal to or in excess of the rejection test standard shall be classed defective. The acceptance test standard shall consist of a warhead assembly which permits no leakage other than that resulting from a .002 inch maximum diameter hole located in the body. Test equipment sensitivity verification shall be performed at one hour maximum intervals of production, and immediately prior to and at the end of acceptance operations for each shift.

4.4.6 Electrical Continuity.-The electrical resistance shall be measured between each socket connector and the corresponding contact at the ogive opening in the finished warhead metal parts assembly. The assembly shall be classed defective if the resistance exceeds the specified requirement.

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4.4.7 Dielectric Strength.-The specified test potential shall be applied between the two conductors, either at the socket connectors or at the contacts of the complete warhead metal parts assembly, and the leakage current measured. The test potential shall then be applied between each conductor (socket connector or contact) and the body, liner and ogive, and the leakage current measured. The warhead assembly shall be classed defective if the leakage current exceeds the specified requirement during any of the tests.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.-The warhead assemblies shall be packaged, packed and marked in accordance with MIL-STD-1169.

5.2 Shipment of Base Connector Caps.-The base connector caps shall be shipped in bulk, loose pack, not to exceed 10 pound packages. Marking of the packaging and packing shall identify the drawing number, drawing revision number and "FOR WARHEAD, HE, M18A1."

6. NOTES

6.1 *Ordering Data.-Procurement documents shall specify* the following:

a. Title, number and date of this specification.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 *Inspection Code Numbers.-The five-digit code numbers* assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (See 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawings numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2. and 6.3.3, should he elect to do so.

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6.3.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

d. The base connector assemblies (9251979) will be supplied as GFM at the time of contract award.

6.3.2 Contractor Designs.-Contractor desgins are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs, if not otherwise covered thereby (See 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract.

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Partial submission of inspection equipment is permissible and encouraged; however, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Sample.-Instructions as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-QA-I.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-MU

Preparing Activity:
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Project Number: 1340-A325

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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